Swing-Up and Stabilization of a Cart-Pole with a Four-Neuron Analog Spiking Network

Transferring evolved sign-activation weights onto low-cost Lu.i hardware

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Abstract

Analog leaky integrate-and-fire (LIF) neurons are attractive for low-power neuromorphic control, but physical implementations introduce noise, device mismatch, finite latency, and limited weight precision. We test whether a minimal analog spiking network can solve a nonlinear control task by implementing a cart-pole controller on Lu.i hardware. The controller is a 3-3-1 feed-forward sign network implemented with four physical Lu.i neurons: three hidden neurons and one output neuron, while the three plant states $(\theta, \dot{\theta}, \dot{x})$ are latency-encoded externally. Weights were optimized offline in a cart-pole simulator using a gradient-free evolutionary strategy and then transferred to hardware by calibrating the Lu.i synaptic potentiometers; biases were applied digitally on an Arduino. In hardware-in-the-loop experiments with a simulated plant, the physical controller swings the pole up from the hanging state in approximately 5 s and stabilizes it near the upright equilibrium. Across repeated hardware trials, the controller succeeded in 98/101 runs, corresponding to a success rate of 97.03%. The stabilized system settles into a small bang-bang limit cycle, consistent with sign-based actuation and analog latency. These results suggest that low-cost analog LIF hardware can implement compact nonlinear feedback controllers, while also highlighting the need for explicit calibration-error and simulation-to-hardware transfer analysis.

1 Introduction

The leaky integrate-and-fire (LIF) model [1] offers an energy-efficient route to neuromorphic computing [2]. We study control on Lu.i, a palm-sized analog PCB implementation of LIF neurons [3]. Unlike simulation, these physical neurons introduce noise, device mismatch, and limited precision. These constraints make cart-pole stabilization a useful benchmark: the system is nonlinear, open-loop unstable, and requires fast closed-loop feedback.

Previous work used near-linear LIF dynamics for LQR-style control [4], often requiring neuron populations to obtain accurate rate-based representations. Here, we instead avoid precise firing-rate estimation. We treat threshold crossings around zero as sign activations and use a compact controller of four physical analog LIF neurons: the three state inputs are latency-encoded directly, so the network comprises three hidden neurons and one output neuron.

1.1 Contributions

This work makes three contributions. First, it demonstrates swing-up and stabilization of a simulated cart-pole using a physical four-neuron Lu.i controller in a hardware-in-the-loop setting. Second, it gives a practical procedure for transferring an evolved sign-activation controller to analog LIF hardware by calibrating potentiometer-set synaptic weights. Third, it evaluates the transferred controller over both swing-up and small-offset stabilization trials, including repeated hardware runs to assess run-to-run repeatability under analog noise.

1.2 Research Questions

We address the following research questions:

(i) Can a four-neuron analog Lu.i network implement a sign-activation controller that swings up and stabilizes a cart-pole in a hardware-in-the-loop setting?

(ii) How reliably does the transferred physical controller perform across repeated trials and different initial pole offsets?

(iii) What calibration information is needed to reproduce the transfer from the evolved simulated network to noisy analog hardware?

2 Theory

- **Cart-Pole.** The plant is the standard simulated cart-pole [5]: a point-mass pole (m, length ℓ) hinged to a cart (M) on a horizontal rail and driven by a horizontal force. The angle θ is measured from the upward vertical, so $\theta = 0$ is the unstable upright equilibrium and $\theta = \pm\pi$ the hanging rest state; the goal is to reach and hold $\theta = 0$, both from a small offset ϵ (stabilization) and from hanging (swing-up). The force takes two levels, $F = \pm F_{\text{max}}$ push left/right (bang-bang control), that is the output of a sign network and keeps training and wiring simple. The controller reads only θ, $\dot{\theta}$ and $\dot{x}$, never the cart position. The point mass dynamics (frictionless rail) are [6]

$$\ddot{x} = \frac{F + m\ell\dot{\theta}^2 \sin\theta - mg\sin\theta\cos\theta}{M + m\sin^2\theta}$$
$$\ddot{\theta} = \frac{(M+m)g\sin\theta - \cos\theta\,(F + m\ell\dot{\theta}^2\sin\theta)}{\ell\,(M + m\sin^2\theta)}$$

- **Lu.i Neurons.** Each neuron is a palm-sized analog leaky integrate-and-fire (LIF) cell [3]: current-based synaptic inputs are integrated on a leaky membrane, and when the membrane crosses its threshold the neuron fires a spike and resets. Its three synapses have potentiometer-set weights with selectable sign (excitatory/inhibitory). Because information is carried in the spike rate, we encode each state variable as an input latency and read the controller's decision from the output neuron's rate. The time between input spikes is computed as $\Delta t = \frac{1}{z_{\text{in}}}$

- **Controller.** The controller is a compact 3-3-1 feed-forward network of these neurons with $\mathrm{sign}(\cdot)$ activation; the output sign sets the force direction, $F = F_{\max} z_{\text{out}}$.
- **Optimization details.** The evolutionary strategy optimized the weight matrices and biases of the sign network,

$$h_j = \mathrm{sign}\left(\sum_{i=1}^{3} W^{(1)}_{ji} z_i + b^{(1)}_j\right), \qquad z_{\text{out}} = \mathrm{sign}\left(\sum_{j=1}^{3} W^{(2)}_{j} h_j + b^{(2)}\right),$$

where $z = (\theta, \dot{\theta}, \dot{x})$ denotes the normalized input state. The optimizer used a population size of 512, mutation scale 0.1, selection rule where the best 32 were chosen, and 128 generations. Each candidate controller was evaluated for 8 episodes of length 60 s, each with slightly varied weights and biases to make it resilient to noise. Initial conditions sampled from $1 + \frac{1}{10}U$ rad, where $U \sim \mathrm{Unif}(-1, 1)$. The best-performing parameter set was then transferred to the Lu.i hardware. The weights and biases were clamped between -1 and $+1$ during training.

Table 1: Final trained sign-network parameters before hardware transfer.

Layer	Input 1	Input 2	Input 3	Bias
Hidden neuron 1	-1	-1	-1	1
Hidden neuron 2	-1	1	1	1
Hidden neuron 3	-1	-1	-1	0
Output neuron	-1	0.39	-1	0

3 Methods

- **Simulation.** The cart-pole equations of motion were integrated in a custom simulator with explicit (forward) RK4 at a fixed step $\Delta t = 1/60$ s over a 60 s time period. One control update is issued per step, giving the analog network a fixed $\approx$16.7 ms window to respond, however in practice, it uses more time, because the signal needs to propagate through the network. The same simulator was used both to obtain the controller offline and to evaluate the hardware-transferred policy in closed loop. We used the following standard constants in our simulation Table 2.
- **Spike encoding.** Communication between neurons is rate-based, so every signal is carried as a spike frequency. Each state variable $z_{\text{in},i}$ is mapped to an input frequency about a fixed offset,

$$\frac{1}{\Delta t_{\text{in},i}} = f_{\text{in},i} = f_{\text{offset},i} + f_{\text{scale},i}\, z_{\text{in},i}.$$

The frequency offsets and gains used for the three input variables are listed in Table 3. Frequencies were clipped to the interval $[f_{\min}, f_{\max}]$ to keep the Lu.i neurons within their reliable operating range. A first-order Taylor expansion of a neuron's frequency transfer

Table 2: Parameters for the numerical cart pole simulation.

Symbol	**Description**	**Value**	**Unit**
Physical parameters			
M	Cart mass	1.0	kg
m	Pole point mass	0.1	kg
ℓ	Pole length	0.5	m
g	Gravitational acceleration	9.81	$\mathrm{m\,s^{-2}}$
Control input			
$F_{\max}$	Max applied horizontal force	8	N
Simulation / integrator settings			
Δt	Integration time step	1/60	s
t_f	Final time	60	s
N	Number of steps ($t_f/\Delta t$)	3600	—
	Integration scheme	RK4	—

Table 3: Latency-encoding parameters for the cart-pole state variables.

State variable	f_{offset}	f_{scale}	$f_{\min}$	$f_{\max}$
θ	10 Hz	3 Hz	7 Hz	13 Hz
$\dot{\theta}$	10 Hz	3 Hz	7 Hz	13 Hz
$\dot{x}$	10 Hz	3 Hz	7 Hz	13 Hz

function about that offset,

$$f_{\text{out}} \approx f_{\text{baseline}} + \sum_i w_i \left(f_{\text{in},i} - f_{\text{offset},i}\right),$$

defines the effective weight w_i as the local sensitivity of the output rate to input i, and the baseline $f_{\text{baseline}} = f_{\text{out}}(f_{\text{offset},i})$ as the output at the operating point, the latter being directly measurable. The binary output follows from thresholding this rate against its baseline,

$$z_{\text{out}} = \text{sign}\big(f_{\text{out}} - f_{\text{baseline}} + f_{\text{bias}}\big) \approx \text{sign}\Big(f_{\text{scale}} \sum_i w_i z_i + f_{\text{bias}}\Big),$$

where f_{out} is sampled by an Arduino every 5 ms. Hidden-layer outputs are re-encoded as frequencies and propagated; the output neuron's decision is converted to the force command and returned to the simulator. On startup the Arduino sets all input spiking rate on the controller to $f_{\text{offset},i}$, it then measures the output spiking rate over a 5 second window. This is then used as f_{baseline}.

- **Controller.** The controller is a 3-3-1 feed-forward network with sign(·) activations [7], mapping the inputs $(\theta, \dot{\theta}, \dot{x})$ to one signed output that is applied as a fixed-magnitude bang-bang force, $F = F_{\max} z_{\text{out}}$ (push left for -1, right for $+1$).

  Its weights and biases were obtained offline in our cart-pole simulator. Because the sign(·) activation is non-differentiable, the parameters were optimized with a gradient-free evolutionary strategy [8] using an energy-based fitness [9] that rewards proximity to the cart-pole's target energy (energy referenced to pivot height, so the upright state corresponds to $E_{\text{target}} = mg\ell$). More precisely, the reward is

  $$E_{\text{target}} = mg\ell, \quad E_{\max} = 2mg\ell, \quad \text{reward} = 1 - \min\left(1, \frac{|E_{\text{pole}} - E_{\text{target}}|}{E_{\max}}\right),$$

  where E_{pole} is the energy of the pole in the simulator.

  Then, the weights are reproduced on hardware: each synaptic weight was set by tuning the corresponding Lu.i potentiometer until the measured weight matched the simulated value. Biases were implemented digitally on the Arduino, as the Lu.i neurons offered no direct mechanism for setting a bias.

- **Evaluation metrics.** A trial was counted as successful if the pole reached the upright neighborhood

  $$|\theta(t)| < \epsilon$$

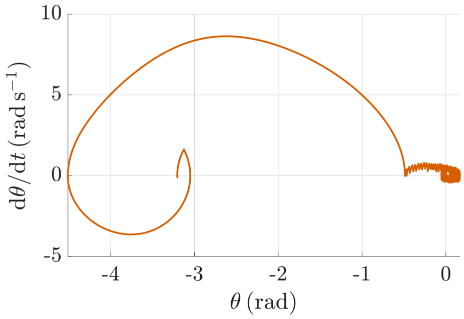
  and remained inside this region for at least $T_{\text{hold}} = 12$ s before the final simulation time $t_f = 60$ s. Unless otherwise stated, we used $\epsilon = 0.02$ rad, matching the settling-radius criterion used in Figure 1d. The settling time t_{stab} was defined as the first time after which the trajectory remained inside this region for the full hold interval:

  $$t_{\text{stab}} = \inf\left\{t : |\theta(\tau)| < \epsilon \text{ for all } \tau \in [t, t + T_{\text{hold}}]\right\}.$$

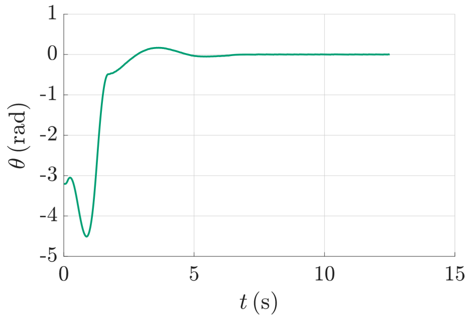
For repeated hardware trials, we report the empirical success rate.

4 Results

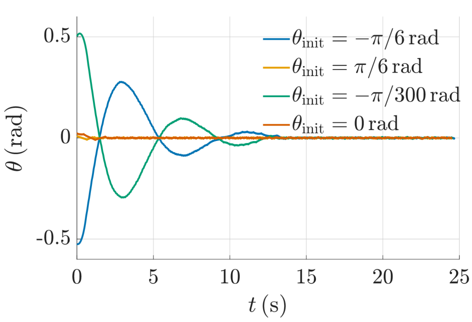
All results below are obtained from the physical Lu.i controller described in the Methods. From the hanging state, the controller swings the pole to upright in $\sim 5\,\mathrm{s}$ (Figure 1a,b) and thereafter holds it within a bang-bang limit cycle of amplitude $\pm 0.1\,\mathrm{rad}$. From an initial offset it returns the pole to vertical for $|\theta_{\mathrm{init}}| \lesssim 0.524\,\mathrm{rad}$, with settling time rising from ~ 0 to $\sim 11\,\mathrm{s}$ as the offset grows (Figure 1c,d). Over 101 repeated hardware runs the controller succeeded in $\frac{98}{101} = 97.03\%$ of trials, indicating its run-to-run repeatability under analog noise.



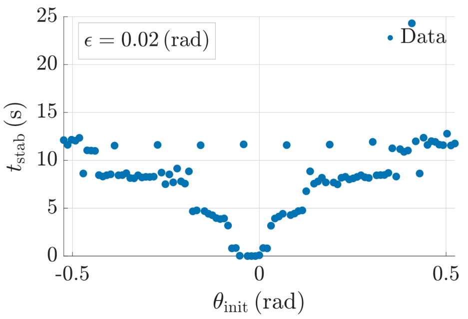
(a) Phase portrait for swing-up from the hanging state. The trajectory approaches the upright equilibrium and settles into a small limit cycle.



(b) Pole angle during swing-up. The pole reaches the upright neighborhood after approximately 5 s.



(c) Stabilization from selected initial offsets near the upright equilibrium.



(d) Settling time as a function of initial pole angle, using the threshold $\epsilon = 0.02\,\mathrm{rad}$.

Figure 1: Closed-loop behavior of the physical Lu.i controller in hardware-in-the-loop experiments: (a)-(b) swing-up from the hanging state and (c)-(d) stabilization from small initial offsets.

5 Discussion

The experiments show that a four-neuron physical Lu.i controller can solve both swing-up and local stabilization in a hardware-in-the-loop cart-pole setting. The phase portrait in Figure 1a is consistent with an energy-pumping strategy during swing-up, followed by capture near the upright equilibrium. The time traces in Figure 1b,c show that the controller can recover from both the hanging state and small initial angular offsets.

The stabilized behavior is not asymptotic convergence to $\theta = 0$. Instead, the controller settles into a small limit cycle around the upright equilibrium. This is expected for a sign-activation controller with fixed-magnitude bang-bang force, because the force command switches discontinuously and cannot apply arbitrarily small corrective inputs. Small spike-counting windows (only last 2 spikes sampled), analog latency, and device noise likely increase the residual jitter and contribute to the scatter in measured settling times.

The results support the feasibility of using low-cost analog LIF hardware for compact nonlinear feedback control. However, the present experiments do not yet isolate the effect of hardware mismatch from the effect of the learned controller itself. A stronger comparison would evaluate the same trained policy in three conditions: ideal simulation, simulation with measured calibration errors, and physical Lu.i hardware in the loop. This would make it possible to quantify the performance loss caused by analog transfer.

The use of sign activations appears useful for robust bang-bang control, but a controlled comparison with smooth activations such as $\tanh(\cdot)$ is needed before claiming a general advantage. Such a comparison should use the same optimizer, plant model, initial-condition distribution, and evaluation metrics for both controller classes.

6 Limitations and Future Work

Several limitations remain.

- First, the cart-pole plant was simulated; therefore, the experiment tests analog-neuron control in a hardware-in-the-loop setting rather than on a fully physical cart-pole. Future work should evaluate the controller on a mechanical cart-pole system with sensor noise, actuator limits, friction, and rail-length constraints.

- Second, the present results do not quantify the difference between the ideal simulated controller and the hardware-transferred Lu.i controller. Future work should report the simulated baseline, the transferred-hardware performance, and the calibration error for each synapse. This would directly measure the simulation-to-hardware performance gap.

- Third, the controller does not use cart position x as an input. Although this keeps the controller minimal, it may allow unbounded cart drift in a physical system with finite rail length. Future evaluations should report the maximum cart displacement,

$$x_{\max} = \max_{t \in [0, t_f]} |x(t)|,$$

  and include rail-bound constraints in the reward function.

- Fourth, some computations are still performed digitally on the Arduino, including bias application and parts of the sign-thresholding pipeline. A more complete analog implementation would move these operations onto the neuromorphic hardware or quantify the fraction of computation performed digitally.

- Fifth, during training, we did not account for potential latency between sensing, and the controller outputting a correcting force. If we had, the bang-bang oscillations would have potentially been reduced.
- Finally, the latency-encoding scheme may not be optimal for noise robustness. Future work should compare alternative encodings, large spike-counting windows, and threshold choices using identical initial conditions and success criteria.

7 Conclusion

We demonstrated that a compact 3-3-1 sign-activation controller implemented with four physical Lu.i neurons can swing up and stabilize a simulated cart-pole in a hardware-in-the-loop setting. The controller uses only three sensed state variables, $(\theta, \dot{\theta}, \dot{x})$, and produces a fixed-magnitude bang-bang force. Across repeated hardware trials, it achieved a success rate of $98/101 = 97.03\%$, while settling into a small limit cycle near the upright equilibrium. These results show that low-cost analog LIF hardware can implement a functional nonlinear feedback policy, but they also indicate that future work should quantify calibration error, simulation-to-hardware transfer loss, cart-position drift, and the contribution of remaining digital computations.

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